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工作经历

博士后研究员	2023.10 - 至今
波士顿大学电子与计算机工程系 Ji-xin Cheng 教授课题组	波士顿
特聘副研究员	2021.7 - 2023.10
北京大学物理学院施可彬教授课题组	北京
博士后	2019.7 - 2021.6
北京大学物理学院施可彬教授课题组	北京

教育背景

北京大学	博士-光学 导师: 施可彬教授	2014 - 2019
南开大学	本科-物理学	2010 - 2014

专业技能

- [光学] 研究方向: 计算成像、非线性光学、全息/数字全息、显微成像、光谱学
技术专长: 复杂光学成像系统的方案设计、光路搭建与调试、系统稳定性与性能优化
应用领域: 活细胞成像、无标记/荧光成像、多模态与超分辨显微成像
- [编程] MATLAB, C/C++, Labview, Python, Bash, $\text{\LaTeX}$
- 仪器控制与自动化: 精通图像/信号采集与控制程序开发, 熟练掌握嵌入式系统软硬件协同设计
- 图像/信号处理: 熟练掌握常用机器视觉和图像/信号处理算法, 具有丰富的相关算法开发与优化经验
- 高新能计算: 掌握基于 GPU 与 Linux 集群的并行计算编程及实时图像处理程序开发
- [软件] Fiji/ImageJ, Mathematica, OriginPro, Adobe AI/PS, Zemax, COMSOL

代表性论文

- (1) Dashan Dong[#], Danchen Jia, Xinyan Teng, Jianpeng Ao, George Abu-Aqil, Biwen Gao, Meng Zhang, Qing Xia, Ji-Xin Cheng[✉]. “ Chem-SIM: SIM-resolved chemical imaging via wide-field mid-infrared photothermal modulation of fluorescence ”. *Nature Communications*, (2026).
- (2) Lige Liu[#], Dashan Dong[#], Zhiwei Long, Wanxue Wei, Chang Sun, Wei Liu, Xiaoshuai Huang, Liangyi Chen, Haizheng Zhong[✉], and Kebin Shi[✉]. “ Single particle fluorescence imaging of perovskite nanocrystal crystallization for illustrating coupled nucleation-and-growth ”. *Nature Communications*, 16.5664, (2025).
- (3) Shuqi Mu[#], Yingtong Shi, Yintong Song, Wei Liu, Wanxue Wei, Qihuang Gong, Dashan Dong[✉], and Kebin Shi. “ Multislice computational model for birefringent scattering ”. *Optica*, 10.1, (2023).
- (4) Shuqi Mu[#], Yu Yang, Juhao Li, Dashan Dong[✉], Ruijun Lan[✉], Kebin Shi. “ Three dimension refractive index characterization for photonic waveguides ”. *Journal of Lightwave Technology*, 40.8, (2022).
- (5) Dashan Dong[#], Xiaoshuai Huang[#], Liuju Li[#], Heng Mao, Yanquan Mo, Guangyi Zhang, Zhe Zhang, Jaiyu Shen, Wei Liu, Zeming Wu, Guanghui Liu, Yanmei Liu, Hong Yang, Qihuang Gong, Kebin Shi[✉], and Liangyi Chen[✉]. “ Super-resolution fluorescence-assisted diffraction computational tomography reveals the three-dimensional landscape of the cellular organelle interactome ”. *Light: Science & Applications*, 9.11, (2020).

- (1) Jianpeng Ao[#], Jiaze Yin[#], Haonan Lin, Guangrui Ding, Youchen Guan, Marzia Savini, Bethany Weinberg, **Dashan Dong**, Qing Xia, Zhongyue Guo, Bowen Liu, Biwen Gao, Ji-Xin Cheng[✉], and Meng C Wang[✉]. “ [FILM: mapping organellar metabolism by mid-infrared photothermal-modulated fluorescence](#) ”. *Nature Methods*, 23.1196-1206, (2026).
- (2) Danchen Jia[#], **Dashan Dong**[#], Tongyu Li, Haonan Zong, Jiabei Zhu, Xinyan Teng, Lei Tian[✉], Ji-Xin Cheng[✉]. “ [Mid-Infrared Photothermal Relaxation Intensity Diffraction Tomography for Video-rate Volumetric Chemical Imaging](#) ”. *Optics Express*, 34.9, (2026).
- (3) Kai Zhang[#], Shuqi Mu[#], **Dashan Dong**, Qihuang Gong, Yue Wang, and Kebin Shi[✉]. “ [Full-component tomographic reconstruction of a dielectric tensor using a multislice computational model](#) ”. *Advanced Imaging*, 3.3, (2026).
- (4) Tongyu Li[#], Yi Shen, **Dashan Dong**, Danchen Jia, Jianpeng Ao, Ji-Xin Cheng, Lei Tian[✉]. “ [Transfer-function approach to substrate-enhanced diffraction tomography](#) ”. *Optica*, 13.4, (2026).
- (5) George Abu-Aqil[#], **Dashan Dong**[#], Jiaze Yin[#], Jianpeng Ao, Hongjian He, Guangrui Ding, Qing Xia, Ji-Xin Cheng[✉]. “ [Bond-Selective Imaging via Vibrational Relaxation Encoded Fluorescence](#) ”. *The Journal of Physical Chemistry Letters*, 16.48, (2025).
- (6) Jia Danchen[#], Huang Steven[#], Tulegenov Dias, **Dashan Dong**, Shvets Gennady[✉], Cheng Ji-Xin[✉]. “ [Metasurface-enhanced infrared photothermal microscopy towards ultrasensitive chemical analysis](#) ”. *Advanced Photonics*, 7.5, (2025).
- (7) Qian Chen[#], Wen Gou[#], Wenqing Lu[#], Jie Li, Yuhong Wei, Haoyu Li, Chengyu Wang, Wei You, Zhengqian Li, **Dashan Dong**, Xiuli Bi, Bin Xiao[✉], Liangyi Chen[✉], Kebin Shi[✉], Junchao Fan[✉], Xiaoshuai Huang[✉]. “ [Fast, three-dimensional, live-cell super-resolution imaging with multiplane structured illumination microscopy](#) ”. *Nature Photonics*, 19.567-576, (2025).
- (8) Jiaze Yin[#], Christian Pfluegl, Chu C. Teng, Rylie Bolarinho, Guo Chen, Xinrui Gong, **Dashan Dong**, Daryoosh Vakhshoori, Ji-Xin Cheng[✉]. “ [Mid-infrared Energy Deposition Spectroscopy](#) ”. *Physical Review Letters*, 134.9, (2025).
- (9) Yue Wang[#], Jingrui Gong, Ning Xu, Shaohui Yan, **Dashan Dong**, Kebin Shi[✉]. “ [Large Field of View and Isotropic Light Sheet Microscopy with Aberration-Free Tunable Foci](#) ”. *Laser & Photonics Reviews*, 19.3, (2025).
- (10) Wenkai Yang[#], Zijian Wang, Jian Xu, **Dashan Dong**, Guiyuan Cao, Han Lin, Baohua Jia, Lige Liu[✉], Kebin Shi[✉]. “ [Ultra-compact computational spectroscopy with a detour-phased planar lens](#) ”. *Light: Advanced Manufacturing*, 5.44, (2024).
- (11) Ziheng Ji[#], Wentao Yu, **Dashan Dong**, Hong Yang, Kaihui Liu, Yun-Feng Xiao, Qihuang Gong, Qinghai Song[✉], Kebin Shi[✉]. “ [High spatial resolution collinear chiral sum-frequency generation microscopy](#) ”. *Advanced Photonics Nexus*, 3.2, (2024).
- (12) Yue Wang[#], **Dashan Dong**, Wenkai Yang, Renxi He, Ming Lei, and Kebin Shi[✉]. “ [Reflective ultrathin light-sheet microscopy with isotropic 3D resolutions](#) ”. *Photonics Research*, 12.2, (2024).
- (13) Peng Liu[#], Jing Shi[#], Danli Sheng[#], Wenqing Lu, Jie Guo, Lei Gao, Xiaoqing Wang, Shaofeng Wu, Yanwen Feng, **Dashan Dong**, Xiaoshuai Huang[✉], and Hongyun Tang[✉]. “ [Mitophrogenesis, a form of mitochondria-specific ectocytosis, regulates sperm mitochondrial quantity and fertility](#) ”. *Nature Cell Biology*, 25.1625-1636, (2023).
- (14) 魏婉雪[#], 何苗, 徐坚, 董大山[✉]. “ [一种大视野快速光学衍射层析成像技术](#) ”. *中国激光*, 50.3, (2023).
- (15) Wenkai Yang[#], Lige Liu[✉], **Dashan Dong**, Yunan Gao, Hong Yang, Qihuang Gong, and Kebin Shi[✉]. “ [In situ three-dimensional observation of perovskite crystallization revealed by two-photon fluorescence imaging](#) ”. *Advanced Optical Materials*, 10.13, (2022).
- (16) Wenkai Yang[#], Lige Liu[#], **Dashan Dong**, Xin Zhang, Han Lin, Yunkun Wang, Hong Yang, Yunan Gao, Haizheng Zhong, Baohua Jia, and Kebin Shi[✉]. “ [Detour-phased perovskite ultrathin planar lens using direct femtosecond laser writing](#) ”. *Photonics Research*, 10.12, (2022).
- (17) Runfeng Li[#], Ruijun Lan[✉], **Dashan Dong**, Hong Yang, and Kebin Shi[✉]. “ [Bessel beam coherent anti-Stokes Raman scattering spectroscopy for turbulent flow diagnosis](#) ”. *Applied Spectroscopy*, 77.3 (2022).

- (18) 穆书奇[#], 董大山, 施可彬[✉]. “ 无标记光学成像技术[🔗] ”. 激光与光电子学进展, 59.12, (2022).
- (19) 李润丰[#], 董大山, 施可彬[✉]. “ 光场调控在相干拉曼散射光谱与成像中的应用 (特邀)[🔗] ”. 光子学报, 51.1, (2022).
- (20) Wei Liu[#], Dashan Dong, Hong Yang, Qihuang Gong, and Kebin Shi[✉]. “ Robust and high-speed rotation control in optical tweezers by using polarization synthesis based on heterodyne interference[🔗] ”. *Opto-Electronic Advances*, 3.8, (2020).
- (21) Dashan Dong[#], Kebin Shi[✉]. “ Solving the missing cone problem by deep learning[🔗] ”. *Advanced Photonics*, 2.2, (2020).
- (22) Shuanglong Liu[#], Bowen Sheng, Xinqiang Wang[✉], Dashan Dong, Ping Wang, Zhaoying Chen, Tao Wang, Xin Rong, Duo Li, Liuyun Yang, Shangfeng Liu, Mo Li, Jian Zhang, Weikun Ge, Kebin Shi, Yuzhen Tong, Bo Shen. “ Molecular beam epitaxy of single-crystalline aluminum film for low threshold ultraviolet plasmonic nanolasers[🔗] ”. *Applied Physics Letters*, 122.23, (2018).
- (23) Wentao Yu[#], Ziheng Ji, Dashan Dong, Xusan Yang, Yunfeng Xiao, Qihuang Gong, Peng Xi[✉], and Kebin Shi[✉]. “ Superresolution deep imaging with hollow Bessel beam STED microscopy[🔗] ”. *Laser & Photonics Reviews*, 10.1, (2016).
- (24) Yonggang Lv[#], Ziheng Ji, Dashan Dong, Kebin Shi[✉], and Qihuang Gong. “ Wide-field vibrational phase contrast imaging based on coherent anti-Stokes Raman scattering holography[🔗] ”. *Chinese Physics Letters*, 32.7, (2015).
- (25) Bin Tsai[#], Wei Liu, Dashan Dong, Kebin Shi, Liangyi Chen, Ning Gao[✉]. “ Phase separation of Mer2 organizes the meiotic loop-axis structure of chromatin during meiosis I[🔗] ”. *bioRxiv*, (2020).

会议论文

- (1) Dashan Dong and Ji-Xin Cheng. “ Super-resolution Chemical Imaging via Structured Illumination Fluorescence-Detected Mid-Infrared Photothermal Microscopy[🔗] ”. *Novel Techniques in Microscopy*, Optica Biophotonics Congress NTu3C.3 (2025).
- (2) Wanxue Wei, Dashan Dong, Kebin Shi. “ Fast optical diffraction tomography microscopy with large field of view and lossless pupil beam combination[🔗] ”. *Ultrafast Nonlinear Imaging and Spectroscopy XII*, Proc. SPIE 1313910, (2024).
- (3) Dashan Dong, Xiaoshuai Huang, Liuju Li, Kebin Shi, and Liangyi Chen. “ Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome[🔗] ”. *Advanced Optical Imaging Technologies III*, SPIE/COS Photonics Asia 115490G, (2020).
- (4) Dashan Dong, Xiaoshuai Huang, Liuju Li, Kebin Shi, and Liangyi Chen. “ Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome[🔗] ”. *Digital Holography and Three-Dimensional Imaging 2020*, Imaging and Applied Optics Congress, Optica Publishing Group HF1G.6, (2020).
- (5) Dashan Dong, Yanhui Cai, Ziheng ji, Hong Yang, Qihuang Gong, and Kebin Shi. “ Tomographic Diffractive Microscopy for Better 3D Imaging[🔗] ”. *14th International Conference on Photonics and Imaging in Biology and Medicine*, Optica Publishing Group W3A.43, (2017).
- (6) Kebin Shi, Dashan Dong, Yanhui Cai, Wei Liu, Chendi Shao. “ High resolution nonlinear imaging based on optical field engineering[🔗] ”. *Ultrafast Nonlinear Imaging and Spectroscopy V*, Proc. SPIE 1038000Z, (2017).

专利

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|---|--------------------------------|
| (1) 一种双模态显微成像系统和方法 [🔗] | 国际专利 PCT/CN2021/071393, (2020) |
| (2) 一种基于缺失型反射镜的离轴全息合束装置及其方法 [🔗] | 2023110715739, (2023) |
| (3) 一种双模态显微成像系统及其成像方法 [🔗] | ZL202110946936.3, (2022) |
| (4) 基于微棱镜的反射式轴向光片荧光显微成像装置及方法 [🔗] | ZL 202110817118.3, (2022) |
| (5) 一种基于贝塞尔 CARS 的湍流光谱测量系统及其测量方法 [🔗] | ZL202110667298.1, (2022) |
| (6) 一种双模态显微成像系统和方法 [🔗] | ZL202010059510.1, (2022) |
| (7) 基于光学外差干涉法的动态柱矢量光场产生装置及其方法 [🔗] | ZL202010493775.2, (2021) |

科研项目

[课题负责人] 国家重点研发计划下设青年科学家课题 (2022YFC3401103)	2022
活细胞超分辨率三维全景成像与解析技术体系研发课题：无标记成像与大数据机器学习	234 万元
[主持] 国家自然科学基金青年科学基金项目 (12004013)	2021
用于活细胞成像的反射式光学衍射层析成像及其应用	24 万元
[共同负责人] 北京大学临床医学 +X 青年专项	2022
基于无标记成像探究脂滴在卵不对称分裂中的作用	10 万元
[主持] 中国博士后科学基金第 68 批面上资助 (2020M680220)	2021
超分辨荧光辅助的光学衍射层析成像及其应用	8 万元
[参与] 国家自然科学基金面上项目 (12174010)	2022
微腔混沌动力学的超高速光学成像	62 万元

会议报告

[口头报告] Optica Biophotonics Congress 2025	2025.4
Super-Resolution Chemical Imaging via Structured Illumination Fluorescence-Detected Mid-Infrared Photothermal Microscopy	美国-圣地亚哥
[墙报] Chemical Imaging 2023, Gordon Research Conference	2023.8
Optical Diffraction Tomography Reveals the 3D Landscape of Living Cells	美国-波士顿
[邀请报告] 第一届先进成像与信息处理会议暨 2023 中国光学学会全息与光信息处理专委会学术年会	2023.7
面向低光毒性的生物光学成像新技术	江西-井冈山
[邀请报告] 第三届全国光子技术论坛	2023.4
基于光场调控的高分辨成像及应用	广东-广州
[邀请报告] 第二届国际计算成像会议 (CITA2022)	2022.10
基于光学衍射层析的活细胞全景成像	上海(线上)
[邀请报告] 第五届光学青年科学家论坛 (OYSS2022)	2022.9
基于光场调控的高分辨成像及应用	福建-福州
[邀请报告] 第二十届全国基础光学与光物理学术讨论会	2021.9
基于四维时空连续性的光学衍射层析成像	吉林-延吉
[邀请报告] 2021 年中国光学学会学术大会	2021.9
基于四维时空连续性的光学衍射层析成像	广东-深圳
[邀请报告] 第四届光学青年科学家论坛 (OYSS2020)	2020.12
光学衍射层析显微成像及其应用	浙江-宁波
[口头报告] 第二届全国光子技术论坛	2020.11
光学衍射层析显微成像及其应用	广东-广州
[邀请报告] 2020 年全国电子显微学学术年会	2020.11
基于光学衍射层析的活细胞全景成像	四川-成都
[口头报告] 第十二届全国光学青年学术论坛	2020.11
光学衍射层析显微成像及其应用	河北-保定
[口头报告] 2020 年亚洲光电子会议 (Photonics Asia 2020)	2020.10
Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional	

	<i>Landscape of Cellular Organelle Interactome</i>	线上
[口头报告]	Digital Holography and Three-Dimensional Imaging, Imaging and Applied Optics Congress <i>Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome</i>	2020.6 线上
[墙报]	International Conference on Photonics and Imaging in Biology and Medicine <i>Tomographic Diffractive Microscopy for Better 3D Imaging</i>	2017.9 江苏-苏州
[墙报]	SPIE Optics + Photonics 2017 <i>Tomographic Diffractive Microscopy for Better 3D Imaging</i>	2017.8 美国-圣迭戈
[口头报告]	中国物理学会 2016 年秋季学术会议 光学衍射断层显微成像	2016.8 北京

学术服务

[审稿人]	Photonics Research, Optics Letters, Optics Express, Advanced Science, APL Photonics, 中国激光, 光学学报	
[程序委员]	国际计算成像会议 (CSOE-CITA), 专题 9: 生物医学与计算成像	2022, 2023
[青年编委]	《中国激光》杂志	2022, 2023
[志愿者]	SPIE/COS 亚洲光电子会议	2016, 2022
[主席]	北京大学 SPIE 学生分会	2017
[副主席]	北京大学 SPIE 学生分会	2016

学位论文

[博士学位论文]	光学衍射层析显微成像的研究	导师: 施可彬教授, 北京大学
[本科毕业论文]	新型矢量光场的特性研究	指导教师: 李勇男教授, 南开大学

荣誉奖项

[2022]	人工微结构和介观物理国家重点实验室甘子钊优秀博士后奖	北京大学
[2022]	现代光学所重要科研进展	北京大学
[2017]	五四奖学金	北京大学
[2017]	优秀科研单项奖	北京大学
[2013]	合展励学金	中国科技馆发展基金会
[2012]	国家励志奖学金	南开大学
[2012]	第三届中国大学生物理学术竞赛-团队一等奖	
[2012]	三好学生	南开大学
[2011]	优秀学生基础学科奖学金	南开大学